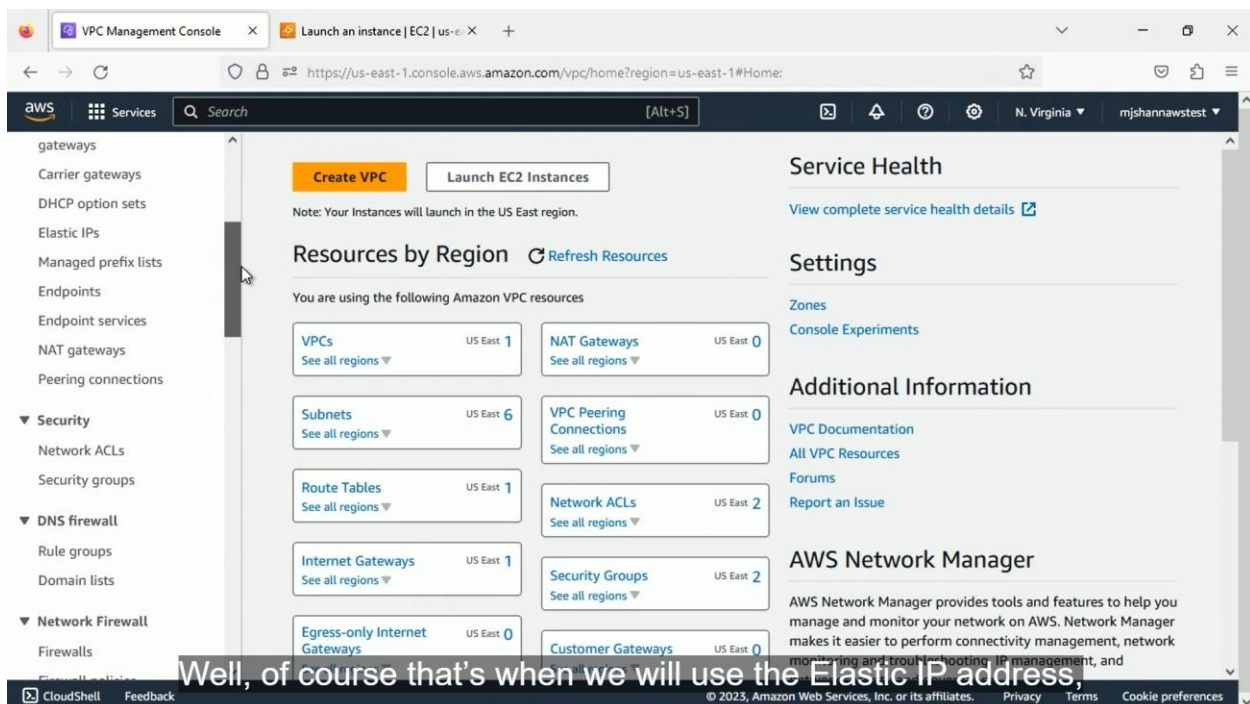


4.1 AWS Network Services

In this course, we will:

- Define the components of a Virtual Private Cloud (VPC)
- Examine VPC addressing
- Survey VPC security
- Learn about VPC peering, endpoints, and the PrivateLink service which leverages endpoints
- Describe Amazon Route 53, edge network services, and various network connectivity options

Exploring Components of a Virtual Private Cloud (VPC)



The screenshot displays the AWS VPC Management Console for the US East-1 region. The left sidebar shows a navigation menu with categories like gateways, DHCP option sets, Elastic IPs, Managed prefix lists, Endpoints, Endpoint services, NAT gateways, Peering connections, Security, DNS firewall, and Network Firewall. The main content area is titled 'Resources by Region' and lists various VPC resources with their counts in US East-1:

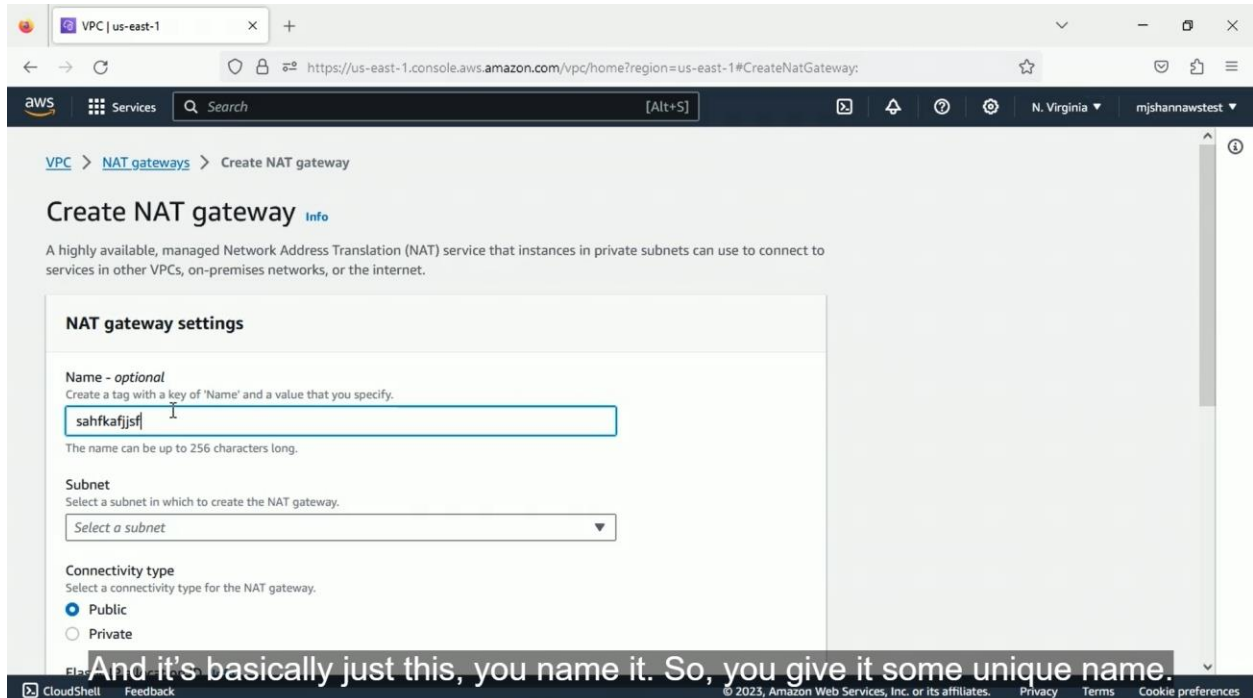
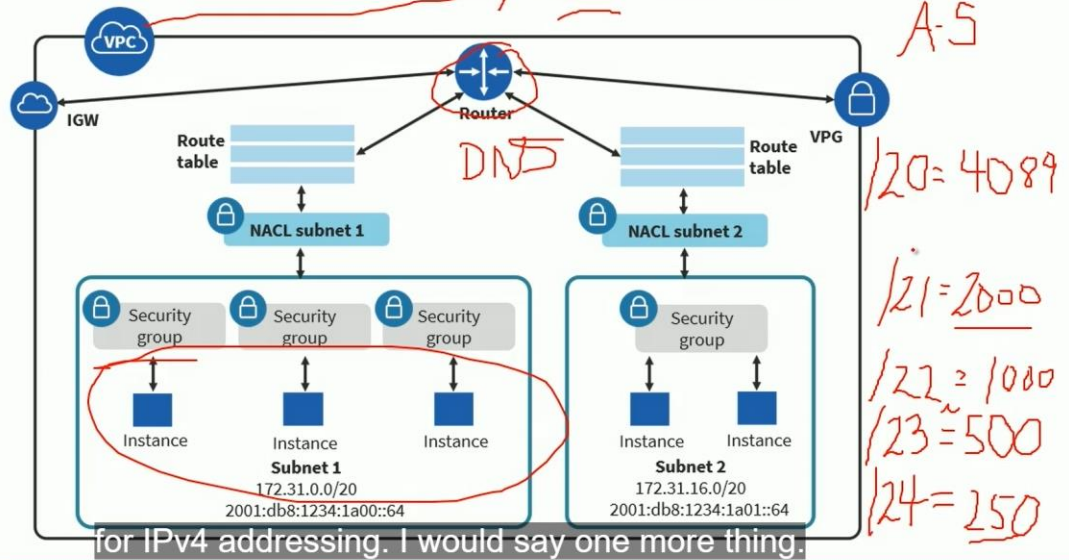
Resource	Count
VPCs	1
NAT Gateways	0
Subnets	6
VPC Peering Connections	0
Route Tables	1
Network ACLs	2
Internet Gateways	1
Security Groups	2
Egress-only Internet Gateways	0
Customer Gateways	0

Additional sections on the right include 'Service Health', 'Settings', 'Additional Information' (with links to VPC Documentation, All VPC Resources, Forums, and Report an Issue), and 'AWS Network Manager'.

Well, of course that's when we will use the Elastic IP address,

Utilizing VPC Addressing

VPC Security Begins with Design



VPC Security

AWS Security Triad

Identity and Access
Management (IAM)

Infrastructure
security

Key Management
Service (KMS)

Most of us are aware of the security triad: confidentiality,

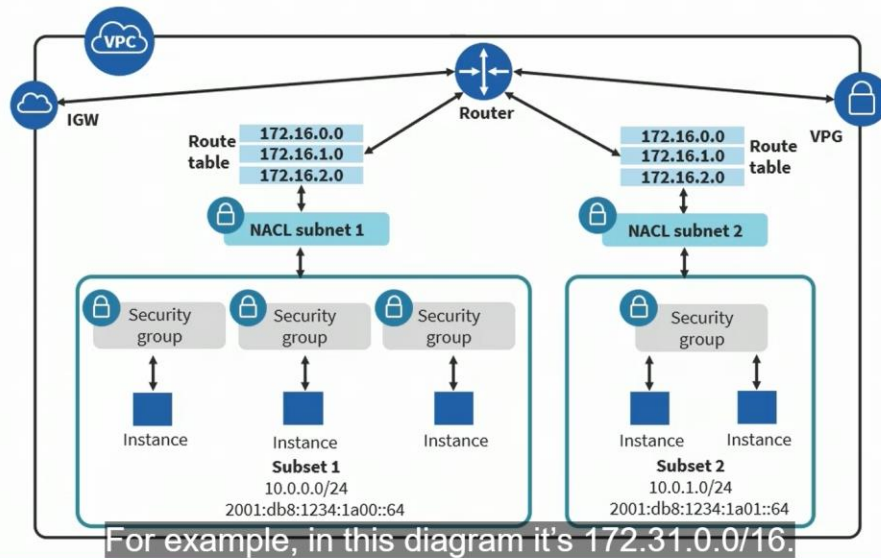
Virtual Private Cloud (VPC) Security

1. Security of the VPC begins with the design of various subnets
2. The next step is to configure the different firewalls at layer 3 and 4 for internal traffic flow
3. The external connectivity options are considered next



Do you use the VPG, virtual private gateway for your Site-to-Site VPN?

VPC Security Begins with Design



Network Access Control Lists (NACLs)

- Allow stateless traffic filtering to all inbound or outbound traffic on a VPC subnet
- Apply to all instances and appliances in the associated subnet
- Typically contain ordered rules to permit or deny traffic:
 - Rules are processed with a numbered order

typically beginning with 100 and then incrementing by fives or tens.

NACLs

- A NACL is agnostic of Transmission Control Protocol (TCP) connections or User Datagram Protocol (UDP)/Internet Control Message Protocol (ICMP) flows
- They are stateless (static) in that the return traffic must be explicitly allowed in the inbound NACL
- These firewalls work together with security groups and can permit or deny traffic before it reaches the interfaces



These firewalls work together as

Security Groups



- Security group (SG) firewalls apply to individual Elastic Compute Cloud (EC2) instances in a subnet
- They operate at the hypervisor level attached to the virtual elastic network interfaces (eth0)
- SGs are layer 3/4 stateful virtual "Allow Only" firewalls
- They have no explicit deny rules like NACLs
- ALL EC2 instances are launched with the default SG unless otherwise designated

the default SG unless otherwise designated.

Security Groups

- An unchanged default SG will allow communication between all resources within the security group AND all outbound traffic – all other traffic is implicitly denied:
 - Return traffic is automatically allowed (with Shield Standard inspection)
- All rules in all applied SGs are evaluated before a decision is made
- There are limits (quotas) on the number of SGs per VPC, on the number of rules you can add to a security group, and on the number of SGs that you can use with an elastic network interface



There are limits or (quotas) on the number of SGs per VPC,

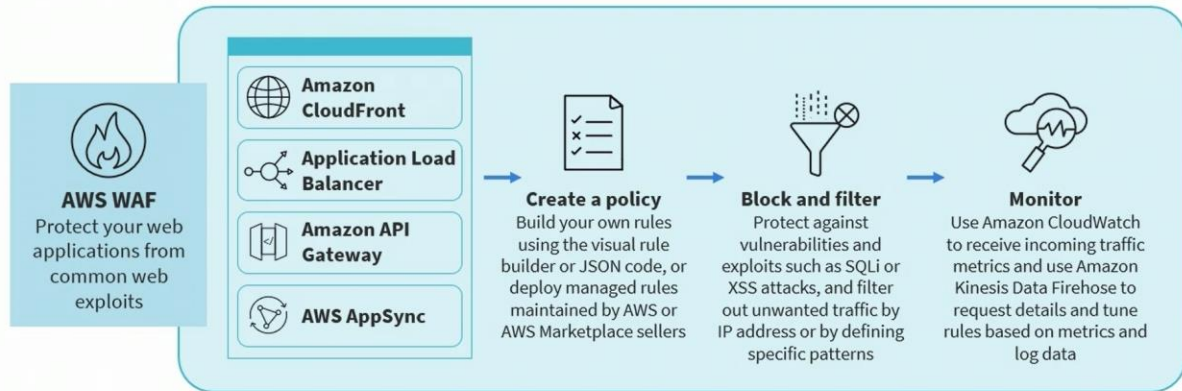
Web Application Firewall (WAF)

- Helps you protect against common web exploits and bots that can affect availability, compromise security, or consume excessive resources
- Is a Layer 7 web access control list (ACL) that uses security rules to control bot traffic and block common attack patterns such as SQL injection, request forgery, or cross-site scripting (XSS)



A WAF is a Layer 7 web access control list, or web ACL,

Web Application Firewall



To use a WAF, you create a policy or more often than not,

VPC Peering

VPC Peering

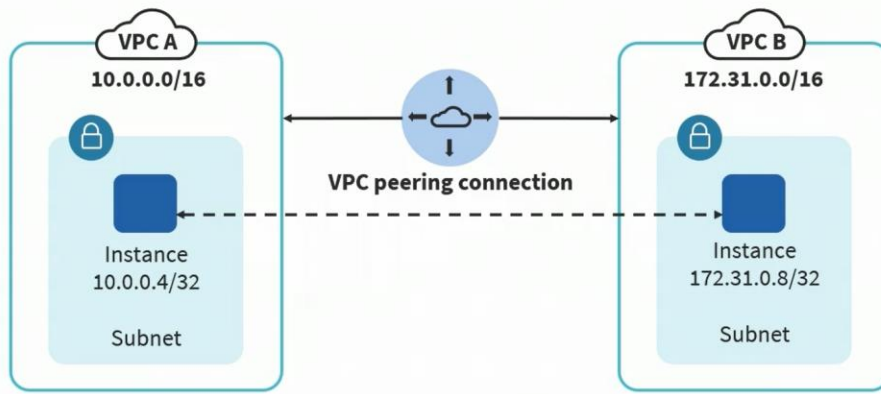
- A VPC peering connection is a networking connection between two VPCs
- Customers can route traffic between them using private IPv4 or IPv6 addresses:
 - Instances in either VPC can communicate with each other as if they are within the same network
- VPC peering connections can be between your own VPCs or with a VPC in another AWS account
- Intra-region or inter-region peering connections are allowed



Intra-region or inter-region peering connections are allowed.

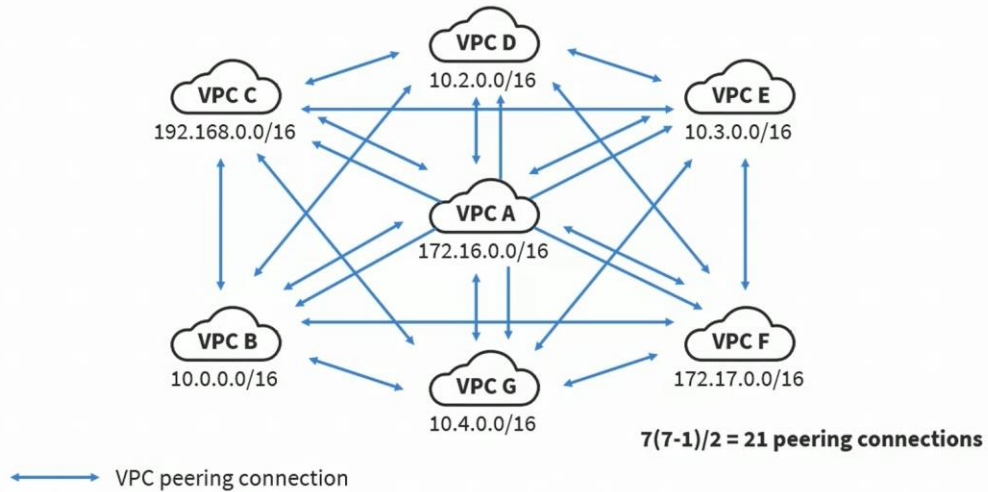
VPC Peering

percipio.com - To exit full screen, press **Esc**



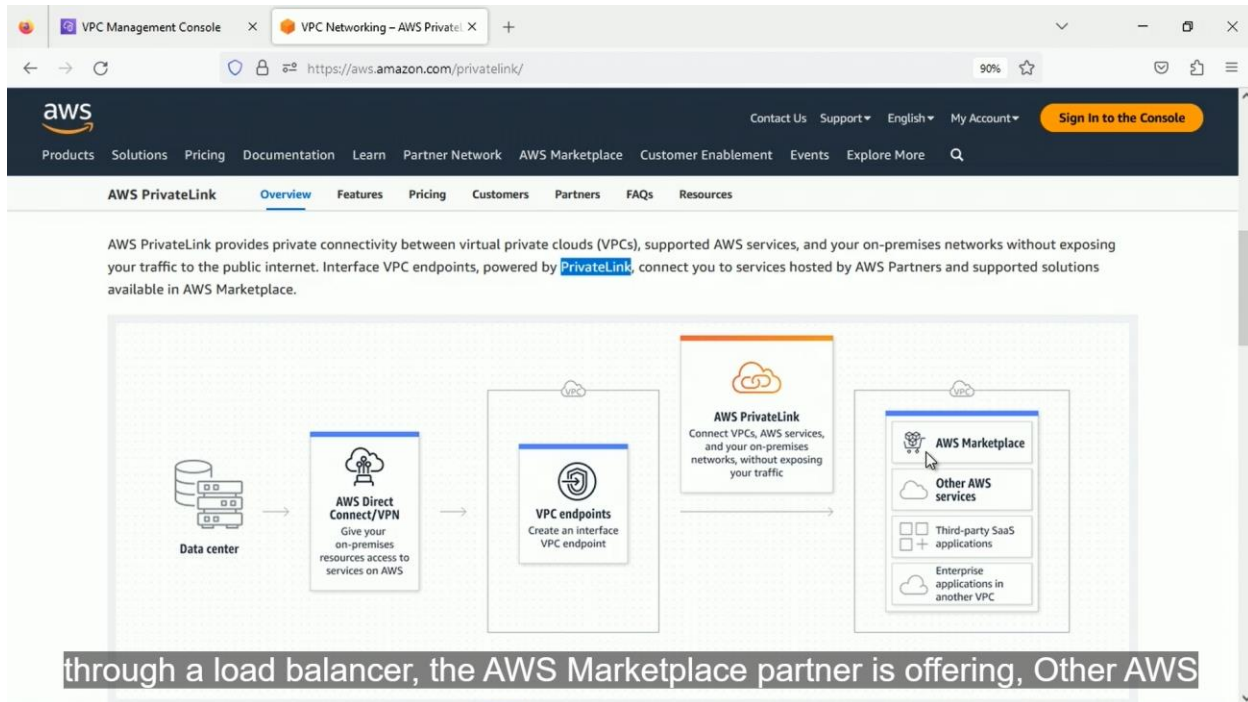
If an individual is an administrator and has

VPC Full-mesh Peering



and lots of VPCs will do full-mesh peering. In this scenario

Comparing Endpoints and Endpoint Services



The screenshot shows the AWS PrivateLink Overview page. The diagram illustrates the following flow:

- Data center** (represented by server icons) connects to **AWS Direct Connect/VPN** (represented by a cloud and server icon).
- AWS Direct Connect/VPN** connects to **VPC endpoints** (represented by a cloud and server icon).
- VPC endpoints** connect to **AWS PrivateLink** (represented by a cloud and server icon).
- AWS PrivateLink** connects to **AWS Marketplace** (represented by a cloud and server icon).
- AWS Marketplace** connects to **Other AWS services** (represented by a cloud and server icon).
- Other AWS services** connects to **Third-party SaaS applications** (represented by a cloud and server icon).
- Third-party SaaS applications** connects to **Enterprise applications in another VPC** (represented by a cloud and server icon).

through a load balancer, the AWS Marketplace partner is offering, Other AWS

Amazon Route 53



Amazon Route 53

- Is a highly available and scalable cloud domain name system (DNS) web service
- Is designed to offer a very dependable and cost-effective method to route end users to Internet applications by translating human-readable domain names to IPv4 and IPv6 addresses

and cost-effective way to route end users to Internet applications by

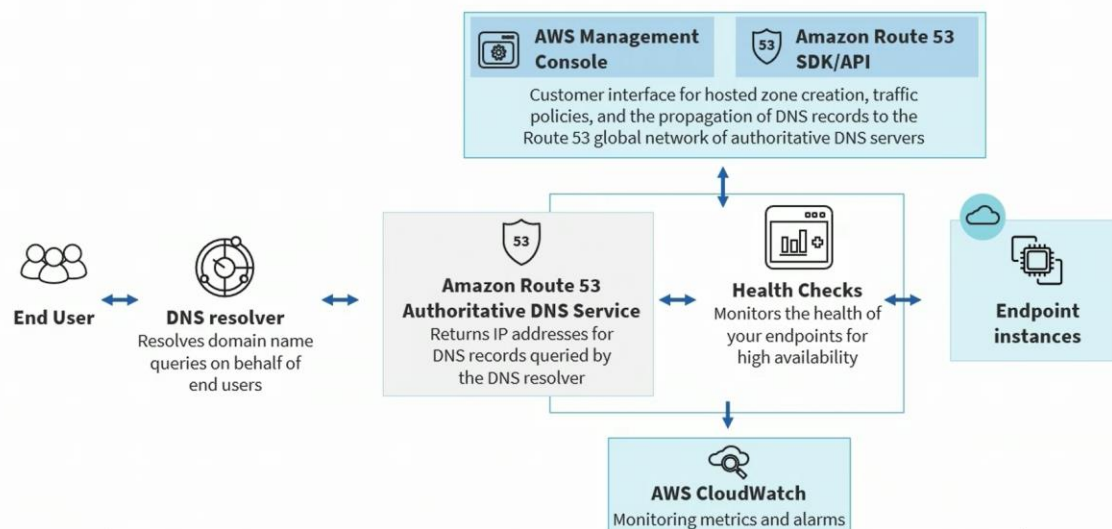
Amazon Route 53

- Effectively connects user requests to AWS services like EC2 instances, Elastic Load Balancing load balancers, and Simple Storage Service (S3) buckets
- Can also be used to route users to infrastructure outside of AWS



It can also be used to route users to infrastructure elements outside of

Amazon Route 53



the end users. Amazon Route 53 has an Authoritative DNS Service,

Edge Network Services

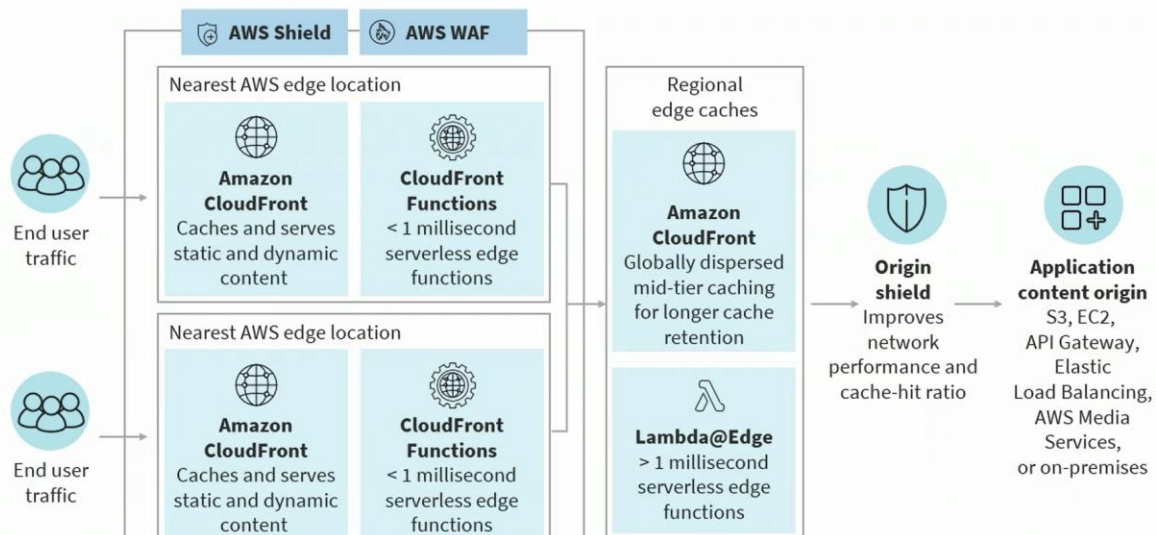
Edge Network Services: Amazon CloudFront

- Is a content delivery network (CDN) service designed for high performance, security, and developer suitability
- Securely delivers data, videos, applications, and application programming interfaces (APIs) to customers globally with low latency and high transfer speeds within a developer-friendly environment
- Is integrated with AWS – both physical locations that are directly connected to the AWS global edge locations and various service endpoints



both physical locations that are directly connected to

Amazon CloudFront



typically in Redis ElastiCache, and serves static and dynamic content.



Edge Network Services: Global Accelerator

- Is a robust networking service that enables customers to expand the availability, performance, and security of their public applications
- Offers two global static public IPs that represent a fixed entry point to application endpoints such as Application Load Balancers, Network Load Balancers, Amazon Elastic Compute Cloud (EC2) instances, and elastic IPs

It offers two global static public IPs that represent

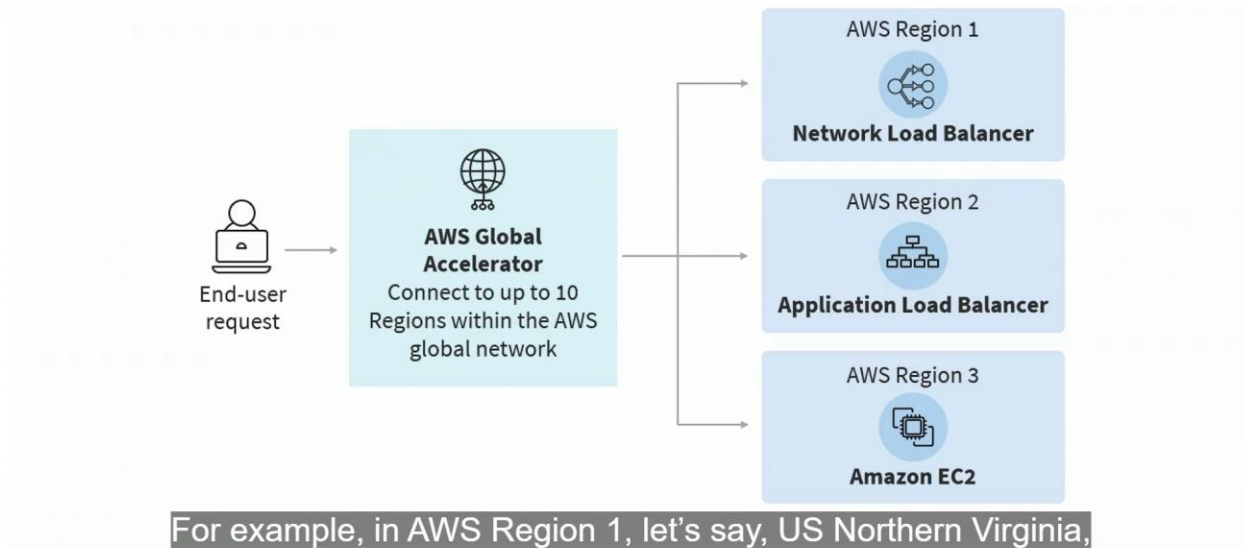
Global Accelerator Use Cases

- Use traffic dials as a global traffic manager to route traffic to the nearest region or attain fast failover across multiple regions
- Accelerate API workloads by up to 60%, leveraging TCP termination at the edge
- Simplify allowlisting in enterprise firewalling and IoT scenarios using global static IPs
- Use custom routing to deterministically route traffic to a fleet of EC2 instances for low-latency gaming and media



And use custom routing techniques to deterministically route traffic to

Global Accelerator



Exploring Network Connectivity Options

The screenshot shows the AWS console 'Create Customer Gateway' page. The 'Details' section on the left contains the following fields:

- Name tag - optional**: A text box with the value 'customer-gateway-01'.
- BGP ASN**: A text box with the value '65000'.
- IP address**: A text box with the value '192.0.2.1'.
- Certificate ARN**: A dropdown menu with the value 'Select certificate ARN'.
- Device - optional**: A text box with the value 'Enter device name'.

On the right side, the 'BGP ASN' section provides information about the Border Gateway Protocol (BGP) Autonomous System Number (ASN). It states that the range of 1 - 2,147,483,647 is supported, with the exception of the following:

- 7224 - Reserved in all Regions
- 9059 - Reserved in the eu-west-1 Region
- 10124 - Reserved in the ap-northeast-1 Region
- 17943 - Reserved in the ap-southeast-1 Region

It also notes that if you don't have a public ASN, you can use a private ASN in the range of 64,512-65,534. The default ASN is 65000. Customer gateways do not support private ASNs in the range 4,200,000,000 to 4,294,967,294.

65,534. Then you would put

1. Which is the default CIDR prefix for the default virtual private cloud in a regional account?
 - 172.31.0.0/16

2. Which AWS service offers private connectivity between VPCs, supported AWS services, and on-premises networks without exposing your traffic to the public internet?
 - PrivateLink
3. Which AWS cloud service is the shortest path to your AWS resources in that while in transit, your network traffic stays on the AWS global network and never touches the public internet?
 - Direct Connect
4. Which AWS VPC element defines the type of subnets you are deploying?
 - Route table
5. Which statement about VPC peering is correct?
 - VPC peering connections must be requested and accepted between two VPCs
6. What is a highly available and scalable cloud DNS service that is designed to offer a very dependable and cost-effective method to route end users to Internet applications by translating human-readable names to IPv4 and IPv6 addresses?
 - Route 53
7. Which is an AWS content delivery network (CDN) service designed for high performance, security, and developer suitability?
 - CloudFront
8. Which type of virtual firewall performs stateless traffic filtering to all inbound or outbound traffic on a VPC subnet?
 - NACL